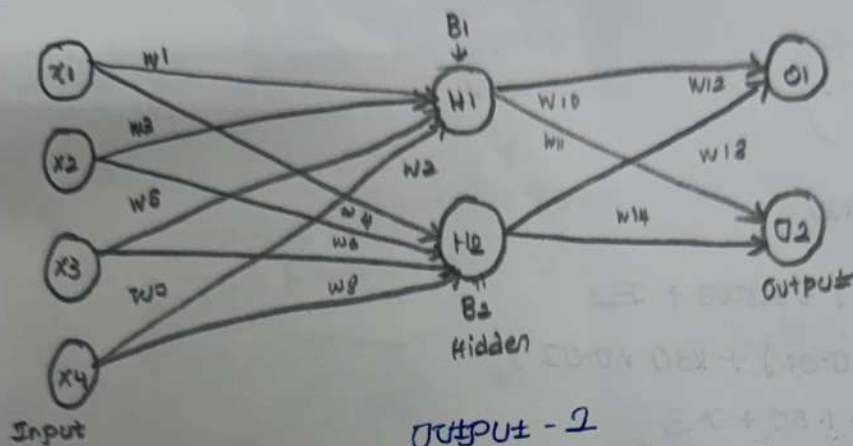


Kavi Varshini's
BCA

Question

The Heart Disease dataset is a Classification dataset used to predict whether a person has heart disease based on medical attributes. The dataset contains features such as Age, Cholesterol, Resting Blood Pressure, and maximum heart rate.



Input - 4

- 1] Age
- 2] Cholesterol
- 3] Resting Blood Pressure
- 4] Maximum Heart Rate

first Row

$$X = [52, 210, 130, 150]$$

Actual Output: ①

Hidden Layer

for H1

$$w_1 = 0.01, w_2 = 0.02, w_3 = 0.01$$

$$w_4 = 0.03, B_1 = 0.5$$

for H2:

$$w_5 = 0.02, w_6 = 0.01, w_7 = 0.07$$

$$w_8 = 0.01, B_2 = 0.3$$

Output - 2

- ① No Heart Disease
- ② Heart Disease

1. Hidden Neuron H1

$$\begin{aligned} Z_1 &= x_1w_1 + x_2w_2 + x_3w_3 + x_4w_4 + B_1 \\ &= (52 \times 0.01) + (210 \times 0.02) + (130 \times 0.01) \\ &= 0.52 + 4.20 + 1.30 + 4.50 + 0.5 \\ \boxed{Z_1 &= 11.02} \end{aligned}$$

Activation

$$\begin{aligned} \sigma(Z_1) &= \frac{1}{1+e^{-Z_1}} \\ \boxed{H_1 &\approx 0.99998 \approx 1} \end{aligned}$$

2. Hidden Neuron H2

$$\begin{aligned} Z_2 &= x_1w_5 + x_2w_6 + x_3w_7 + x_4w_8 + B_2 \\ &= (52 \times 0.02) + (210 \times 0.01) + (130 \times 0.02) \\ &= 1.04 + 2.10 + 2.60 + 1.50 + 0.3 \\ \boxed{Z_2 &= 7.54} \end{aligned}$$

Activation

$$\begin{aligned} \sigma(Z_2) &= \frac{1}{1+e^{-Z_2}} \\ \boxed{H_2 &\approx 0.99947 \approx 1} \end{aligned}$$

Output Layer

$O_1 \rightarrow$ No Heart Disease

- $WH_{1,01} = 0.4$
- $WH_{2,01} = 0.2$
- $BO_1 = 0.1$

$O_2 \rightarrow$ Heart Disease

- $WH_{1,02} = 0.3$
- $WH_{2,02} = 0.8$
- $BO_2 = 0.1$

3] Output for No Heart Disease

$$\begin{aligned} I_{\text{NoHD}} &= H_1(0.4) + H_2(0.2) + 0.1 \\ &= (0.99998 \times 0.4) + (0.99947 \times 0.2) + 0.1 \\ &= 0.39999 + 0.19989 + 0.1 \\ I_{\text{NoHD}} &\approx 0.69988 \end{aligned}$$

4] Output for Heart Diseases

$$\begin{aligned} I_{\text{HD}} &= H_1(0.3) + H_2(0.8) + 0.1 \\ &= (0.99998 \times 0.3) + (0.99947 \times 0.8) + 0.1 \\ &= 0.29999 + 0.79958 + 0.1 \\ &= \boxed{I_{\text{HD}} \approx 1.19957} \end{aligned}$$

$$O_1 = 0.69988$$

$$O_2 = 1.19957$$

$$\boxed{1.19957 > 0.69988}$$

So, O_2 has the highest value

Predicted Class: Heart Disease (1)